Example :

i) Various heavy metals (Ag^{+}, Pb^{2+}, Hg^{2+}) can non-competitively inhibit various enzymes. For example, urease is highly sensitive to all these heavy metal ions.

ii) H_2S and cyanide strongly inhibit the action of iron containing enzymes like cytochrome oxidase.

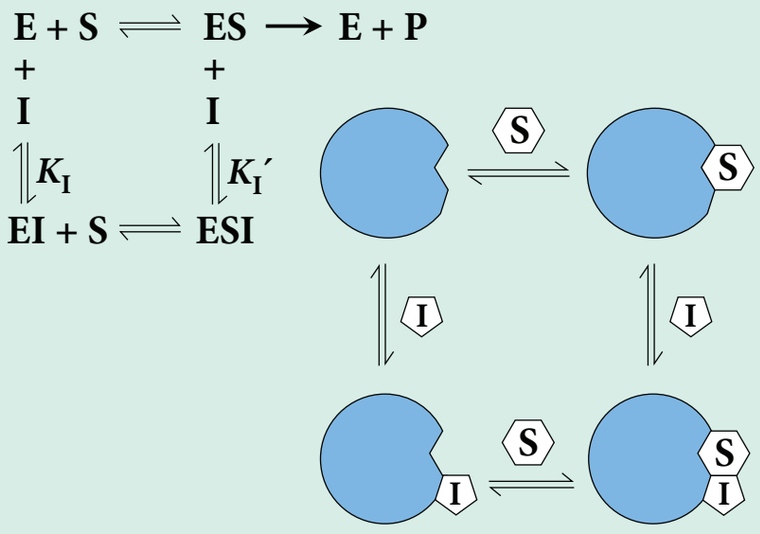


Figure 9.10 Non-competitive inhibition

The action of a non-competitive inhibitor is given as follows (Figure 9.10)

The noncompetitive inhibitor gives two K_i values as follows.

$$K_i = \frac{([E]\ [I])}{([EI])}$$

$$K_i = \frac{([ES]\ [I])}{([ESI])}$$

9.3.1.3. Uncompetitive inhibition :

It is also a type of reversible inhibition in which the inhibitor does not have any affinity for free enzyme, but binds only with the enzyme-substrate complex.

Uncompetitive inhibitor decreases both K_m and V_{max} values of the enzyme.

Example : Inhibition of placental alkaline phosphatase by phenyl alanine is an example of uncompetitive inhibition.

The action of uncompetitive inhibitor may be given as follows (Figure 9.11)

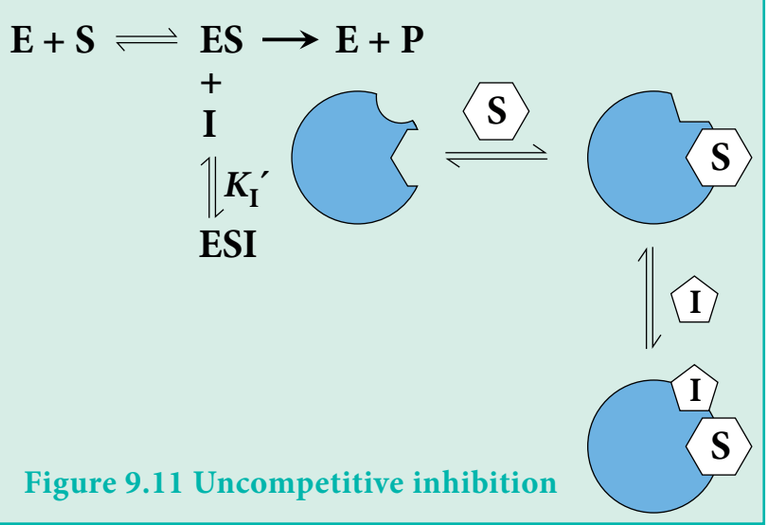


Figure 9.11 Uncompetitive inhibition

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